Tutorial 11

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(1). Lemma 2:-

If language L is recognized by a PDA, then there exists a CFG that generates L.

proof:-

Assume L is recognized by ~~PDA P~~ = PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$.

Convert PDA to accept by empty stack with standard form transitions.

construct CFG $G = (V, \Sigma, R, S)$:

for every pair of ~~step~~ states p, q in Q create variable A_pq.

start symbol $S = A\_q_0q_f$

if $\delta(p, a, x) = (q, \epsilon)$, add rule:

$A\_pq \rightarrow a$

if $\delta(p, a, X) = (r, YZ)$, then

for every state s:

$A\_pq \rightarrow a \; A\_rs \; A\_sq$

Thus every PDA computation corresponds to a CFG derivation.

Hence proved.

(2). (1). $L = \{0^n 1^n \mid n \geq 0\}$

Assume regular. Let pumping length = p

choose $w = 0^p 1^p$

y contains only 0s

pumping changes number of 0s but not 1s.

contradiction

Therefore not regular //

(2). $L = \{w \in \{0, 1\}^* \mid$ equal number of 0s and 1s $\}$

choose $w = 0^p 1^p$

pumping breaks balance

NOT regular. //

(3). $L = \{1^{n^2} \mid n \geq 0\}$

assume L is regular.

Let the pumping length be p

$w = 1^{p^2}$

since $|w| \geq p$, by the pumping lemma we can write; $w = xyz$

where y contains only 1s & $|y| > 0$

pumping y increases the number of 1s by a fixed amount

the new length becomes

$p^2 + k \quad (k > 1)$

This length is not a perfect square, so the pumped string is not of the form 1^{n^2}

There for;

contradiction $\rightarrow$ L is not regular //

3 (1). $L = \{a^n b^n c^n \mid n \geq 0\}$

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choose $w = a^p b^p c^p$

pumping affects at most two symbol type.

cannot maintain equality of three counts.

NOT context-free //

(2). $L = \{ww \mid w \in \{0,1\}^*\}$

choose $w = 0^p 1^p 0^p 1^p$

pumping breaks identical halves property.

NOT context-free. //

(3). $L = \{a^i b^j c^k \mid 0 \leq i \leq j \leq k\}$

choose $w = a^p b^p c^p$

pumping breaks inequality conditions.

NOT context-free //